

A Smart Device for Monitoring Porosity and Moisture Levels of Various Soils

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ABSTRACT ORIGINAL

The sustainability of agriculture may be threatened by continuous cropping if soil nutrients are not properly measured and supplied. Measuring the nutrients in the soil is crucial for healthy plant development and efficient fertilization. Determining the amount of nutrients in the soil is important for formulated fertilization. Recommendations for nutrient requirements and fertilization tailored to the individual site come next. Plant development requires the three essential nutrients: potassium, phosphorus, and nitrogen. The present study involves the development of an electrochemical sensor capable of differentiating various soil nutrients, including N, P, and K. The sensor operates by absorbing ions from the soil's aqueous solution. The integrated circuits of the sensor and Arduino Microcontroller are designed to detect and pinpoint any deficiencies in the soil's components. It has been noted that the suggested algorithm performs better overall than alternative techniques regarding F-Score, recall, accuracy, and precision. With just a little amount of training, the farmer may effortlessly manage his field soil data in the cloud service by use of the IoT SNA-CR gadget on his mobile Smartphone. It also boosts production and lowers the cost of balancing the minerals in the soil. © 2024 IEEE.